COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010 / Data Structures and Algorithms	**SECTION:**	CPE21S3
DATE PERFORMED:	08/25/2026	**DATE SUBMITTED:**	08/25/2026
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PRELIM EXAM **PRELIM EXAM**

1. PROCEDURE OUTPUT

Palindrome	Output	Remark
101	---Queue Linked List--- 1 1 1 0 1 1 Front of the queue: 1 Back of the queue: 1 ---Stack Array--- Enter input: 101 The input is a palindrome. Palindrome: 1 0 1	Success.
1013003001	---Queue Linked List--- 1 1 1 0 1 1 1 3 1 0 1 0 1 3 1 0 1 0 1 1 Front of the queue: 1 Back of the queue: 1 ---Stack Array--- Enter input: 1013003001 The input is a palindrome. Palindrome: 1013003001	Success.

88	---Queue Linked List--- 8 8 8 8 Front of the queue: 8 Back of the queue: 8 ---Stack Array--- Enter input: 88 The input is a palindrome. Palindrome: 88	Success.
0	00000000000000000000 00000000000000000000 00000000000000000000 00000000000000000000 00000000000000000000	Failed.
202	---Queue Linked List--- 2 2 2 0 2 2 Front of the queue: 2 Back of the queue: 2 ---Stack Array--- Enter input: 202 The input is a palindrome. Palindrome: 202	Failed.

Function:

prelimStackArr.h (palindromeChecker):

```
//palindrome checker:
bool palindromeChecker(int open, int close){
    return (open == '1' && close == '1');
}
```

```
bool palindromeChecker(std::string input){
    arrStack stack;
    for (int p: input){
        if (p == 1){
            stack.push(p);
        }
        else if (p == 1){
            if (stack.isEmpty())
                return false;
            int top = stack.pop();
            if(!palindromeChecker(top, p))
                return false;
        }
    }
    return stack.isEmpty();
}
```

Analysis:
I created two palindrome checkers for the stack array. In the first palindrome checker, it will return if both have the same number in the palindrome, I can also edit it if it's a different number. In the second palindrome checker, it needs an input which is in the prelim.cpp (main file) then it will detect in the class then in the for loop, it will detect the for loop to push and if else to pop and return and false, it will return to the stack isEmpty.

prelim.cpp

```
if(palindromeChecker(input)){
    std::cout << "The input is a palindrome." << std::endl;
}else{
    std::cout << "The input is not a palindrome." << std::endl;
}
```

Analysis:
In the palindrome checker it needs an input to check if it's a palindrome or not.

2. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **N/A**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **N/A**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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